

Practical-1

Practical lab

Assignment

The screenshot displays the CODETANTRA online IDE interface. The left sidebar shows the file explorer with a file named 'calculate...'. The main editor area contains a Python script for calculating momentum. The script takes two inputs, calculates the momentum using the formula $p = m \times v$, and prints the result rounded to two decimal places. The right sidebar shows the test results, indicating that 2 out of 2 shown test cases passed and 2 out of 2 hidden test cases passed. The test cases show expected and actual outputs for mass and velocity inputs.

1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 m=float(input())
2 v=float(input())
3 p=m*v
4 print(f"{p:.2f}kgm/s")
```

Test Results:

- Average time: 0.015 s (15.25 ms)
- Maximum time: 0.018 s (18.00 ms)
- 2 out of 2 shown test case(s) passed
- 2 out of 2 hidden test case(s) passed

Test case 1 (18 ms):

Expected output	Actual output
5.0	5.0
18.0	18.0
50.00kgm/s	50.00kgm/s

Test case 2 (14 ms):

Terminal | Test cases

< Prev | Reset | Submit | Next >

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1.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:
The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2 n=int(input())
3 if n>1 and n<=9:
4     print(f"{(n*n)}")
5 elif n>9 and n<99:
6     print(f"{math.sqrt(n):.2f}")
7 elif n>99 and n<999:
8     print(f"{math.cbrt(n):.2f}")
9 else:
10    print("Invalid")
```

Average time
0.008 s
8.29 ms

Maximum time
0.013 s
13.00 ms

4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1
Expected output
9
81

Actual output
9
81

Test case 2

Test case 3

TerminalTest cases

PrevResetSubmitNext

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:
An integer representing the number of days between the given dates.

Note:
The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```
1 from datetime import date
2 from datetime import date
3 def days_between_dates(date_str1,date_str2):
4     year1, month1, day1=map(int,date_str1.split('-'))
5     date1=date(year1, month1,day1)
6     year2,month2,day2=map(int,date_str2.split('-'))
7     date2=date(year2,month2,day2)
8     return abs((date2-date1).days)
9
10 input_date1=input().strip()
11 input_date2=input().strip()
12 print(days_between_dates(input_date1,input_date2))
13
14
15
```

Average time
0.015 s
15.00 ms

Maximum time
0.016 s
16.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1
Expected output
2014-07-02
2014-11-02
123

Actual output
2014-07-02
2014-11-02
123

Test case 2

TerminalTest cases

PrevResetSubmitNext

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1.1.4. Reverse a Number01:04

Write a Python program to reverse the digits of the number and print the reversed number.
Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 #write your code here...
2 n=int(input())
3 rev=str(n[::-1])
4 print(rev)
```

Average time0.008 s7.80 msMaximum time0.010 s10.00 ms

2 out of 2 shown test case(s) passed3 out of 3 hidden test case(s) passed

Test case 110 msDebug

Expected output53677835Actual output53677835

Test case 27 ms

TerminalTest cases

< PrevResetSubmitNext >

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1.1.5. Multiplication Table02:10

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.
Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:
<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 #write your code here...
2 n=int(input())
3 for i in range(1,11):
4     print(f'{n} x {i} = {n*i}')
5
6
```

Average time0.011 s10.75 msMaximum time0.014 s14.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 114 msDebug

Expected output88 x 1 = 88 x 2 = 168 x 3 = 248 x 4 = 328 x 5 = 40Actual output888 x 1 = 88 x 2 = 168 x 3 = 248 x 4 = 328 x 5 = 40

TerminalTest cases

< PrevResetSubmitNext >

Lab Assignment

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1.2.1. Pass or Fail

0.011 s

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

Submit

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

```
def Grades(x):  
    if(x > 75):  
        print("Grade: Distinction")  
    elif (75 >= x >= 60):  
        print("Grade: First Division")  
    elif (60 >= x >= 50):  
        print("Grade: Second Division")  
    elif (50 >= x >= 40):  
        print("Grade: Third Division")  
    else:  
        print("Fail")  
  
n = int(input())  
marks = list(map(int, input().split()))
```

Average time

0.013 s

12.50 ms

Maximum time

0.015 s

15.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1

15 ms

Debug

Expected output

4

55 65 78 89

Actual output

4

55 65 78 89

Aggregate Percentage: 71.75

Grade: First-Division

Test case 2

4 ms

Terminal

Test cases

Prev

Reset

Submit

Next

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1.2.2. Fibonacci Series

0.024 s

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

Submit

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

```
def fibonacci(n):  
    # write the code..  
    if n<=1:  
        return n  
    else:  
        return fibonacci(n-1)+fibonacci(n-2)  
    return fibonacci(  
  
n = int(input())  
for i in range(1, n + 1):  
    print(fibonacci(i), end=" ")  
    
```

Average time

0.024 s

23.50 ms

Maximum time

0.063 s

63.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

40 ms

Debug

Expected output

5

0 1 1 2 3

Actual output

5

0 1 1 2 3

Test case 2

6 ms

Terminal

Test cases

Prev

Reset

Submit

Next

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1.2.3. Pattern - 1

0.057

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

+

rightangl...

Submit

Debugger

1n=int(input(""))

2for i in range(1,n+1):

3for j in range(1,i+1):

4print("*",end=" ")

5print()

Average time0.008 s

Maximum time0.010 s

8.17 ms10.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 110 msDebug

Expected output

Actual output

5

1 *

1 * 1 *

1 * 2 * 1 *

1 * 3 * 2 * 1 *

1 * 4 * 3 * 2 * 1 *

Terminal

Test cases

< Prev

Reset

Submit

Next >

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1.2.4. Pattern - 2

0.134

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

+

numberP...

Submit

Debugger

1n=int(input())

2for i in range(1,n+1):

3for j in range(1,i+1):

4print(j,end=" ")

5print()

6

7

8

Average time0.009 s

Maximum time0.010 s

8.76 ms10.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 110 msDebug

Expected output

Actual output

5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Terminal

Test cases

< Prev

Reset

Submit

Next >